Supplementary Material: OCV and Aging Models

Shida Jiang, Shengyu Tao, Vincent Molina, Junzhe Shi, and Scott Moura

1 Supplementary OCV and Aging Models

This supplementary material provides the open-circuit-voltage (OCV) functions, baseline aging models, and parameter values used in the simulation study. These models are used only to evaluate the proposed control strategy in long-term simulations; they are not embedded in the controller. Thus, the proposed control strategy remains model-agnostic with respect to battery degradation. Throughout this supplement, SOC and SOH are expressed as fractions, and ΔSOH denotes a positive fractional capacity loss.

1.1 OCV–SOC Relationships

Two lithium-ion chemistries are considered in the simulation study: lithium iron phosphate (LFP) and lithium manganese oxide (LMO). The LFP OCV–SOC relationship is adopted from [1]:

$$OCV_{\mathrm{LFP}} = -0.5863\exp(-21.9\,SOC) + 3.414 + 0.1102\,SOC - 0.1718\,\exp\left(\frac{-0.008}{1-SOC}\right), \tag{1}$$

where $SOC \in [0,1]$. The LMO OCV–SOC relationship is adopted from [2]:

$$OCV_{\mathrm{LMO}} = 3.875 - 0.335\big(-\ln(SOC)\big)^{0.653} - 0.5332\,SOC + 0.8315\exp\big(0.6(SOC-1)\big), \tag{2}$$

where $SOC \in [0,1]$.

1.2 Baseline LFP Aging Model

For LFP cells, the baseline degradation model is adopted from [3]. Over an interval in which temperature, C-rate, and average SOC are approximated as constant, the capacity loss is

$$\begin{aligned}\Delta SOH_{\mathrm{LFP,base}} = \frac{1}{Q_{\mathrm{no}}}\Big[&(aT^2+bT+c)\exp\big((dT+e)I^{no}\big)\,\Delta AH \\ &+ f\exp\big(g\overline{SOC}+h/T\big)\left(\sqrt{t_1}-\sqrt{t_0}\right)\Big].\end{aligned} \tag{3}$$

Here T is the temperature in Kelvin, I^{no} is the C-rate, Q_{no} is the nominal capacity, and t_0 and t_1 are the initial and final calendar times in days. The quantity ΔAH is the absolute Ah throughput over the interval, so both charging and discharging contribute to cycling aging. During relaxation, $\Delta AH = 0$ and only the calendar-aging term remains.

1.3 Baseline LMO Aging Model

For LMO cells, the baseline aging model is adopted from [4]. Let f_d denote the cumulative linearized degradation state immediately before an interval. The corresponding finite capacity-loss increment is

$$\Delta SOH_{\text{LMO,base}} = F(f_d + \Delta f_d) - F(f_d), \quad F(x) = 1 - \alpha_{\text{sei}} e^{-\beta_{\text{sei}} x} - (1 - \alpha_{\text{sei}}) e^{-x}. \tag{4}$$

For a cycling interval, the linearized degradation increment is

$$\Delta f_{d,\text{cyc}} = \frac{n}{k_{\delta 1} \delta^{k_{\delta 2}} + k_{\delta 3}} \exp\left[k_\sigma (\sigma - \sigma_{\text{ref}}) + \frac{k_T T_{\text{ref}} (T - T_{\text{ref}})}{T} \right], \tag{5}$$

where δ is the SOC swing, σ is the average SOC, and $n = 1$ or 0.5 for a full or half cycle, respectively. For a calendar-aging interval,

$$\Delta f_{d,\text{cal}} = k_t \Delta t \exp\left[k_\sigma (\sigma - \sigma_{\text{ref}}) + \frac{k_T T_{\text{ref}} (T - T_{\text{ref}})}{T} \right], \tag{6}$$

where Δt is in seconds. In the stage-discretized simulation, the cycling expression is evaluated over each modeled charging stage with $n = 0.5$.

1.4 Model Parameters

The parameters used in the LFP and LMO baseline aging models are listed in Table 1.

Table 1: Parameters of the baseline battery aging models.

Parameter	**Value**	**Parameter**	**Value**
a	2.0916×10^{-8}	b	-1.2179×10^{-5}
c	1.80×10^{-3}	d	-1.7082×10^{-6}
e	5.56×10^{-2}	f	5.9808×10^{6}
g	6.898×10^{-1}	h	-6.4647×10^{3}
α_{sei}	5.75×10^{-2}	β_{sei}	121
$k_{\delta 1}$	1.40×10^{5}	$k_{\delta 2}$	-5.01×10^{-1}
$k_{\delta 3}$	-1.23×10^{5}	k_σ	1.04
σ_{ref}	0.50	k_T	6.93×10^{-2}
T_{ref}	$298.15\,\text{K}$	k_t	$4.14 \times 10^{-10}\,\text{s}^{-1}$

1.5 Pack-Level Cell-to-Cell Heterogeneity and Knee-Point Acceleration

The baseline models do not explicitly represent persistent cell-to-cell variation or accelerated late-life degradation. For either chemistry, the short-term degradation of cell i is therefore augmented as

$$\Delta SOH_i = \gamma_i \left[1 + a_{\text{knee}} \max\{SOH_{\text{knee}} - SOH_i,\, 0\}\right] \Delta SOH_{\text{base}}, \tag{7}$$

where $\gamma_i \sim \mathcal{N}(1, \sigma_\gamma^2)$ is sampled once for each cell and held constant over time. Unless otherwise stated, $\sigma_\gamma = 0.10$, $SOH_{\text{knee}} = 0.75$, and $a_{\text{knee}} = 25$.

References

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[3] J. Nájera, J. R. Arribas, R. M. de Castro, and C. S. Núñez, "Semi-empirical ageing model for LFP and NMC Li-ion battery chemistries," *Journal of Energy Storage*, vol. 72, p. 108016, 2023.

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